

Synthesis Report: A Closed-form Solution to Photorealistic Image Stylization

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Abstract

This report synthesizes the paper “A Closed-form Solution to Photorealistic Image Stylization” by Li et al.[4] and describes an implementation of its core methods via a Jupyter Notebook. The paper introduces a two-stage framework that combines a modified Whitening and Coloring Transform (PhotoWCT) with a graph-based smoothing step formulated as a closed-form optimization problem. Our experiments, performed on novel content/style pairs randomly selected from online searches, illustrate both the strengths and limitations of the approach.

1 Introduction

Photorealistic image stylization seeks to transfer the style of a reference image onto a content image while preserving a natural, photographic appearance. Traditional neural style transfer techniques, such as those proposed by Gatys et al. [1, 2], use iterative optimization to produce artistic renderings, but often at the cost of structural distortions. Subsequent methods (e.g., Luan et al. [5]) have introduced regularization strategies to reduce these artifacts; however, these approaches remain computationally expensive and may still yield noticeable inconsistencies.

In contrast, Li et al. [4] propose a novel closed-form solution that decomposes the style transfer task into two steps: a stylization stage and a smoothing stage. Their method, termed *PhotoWCT*, leverages a modified Whitening and Coloring Transform (WCT) along with graph-based smoothing to achieve high-quality photorealistic stylizations while significantly reducing processing time. This report details the key contributions, optimization techniques, and experimental evaluations presented in the paper. (Appendix A, Figure 2, shows how our method compares to prior approaches such as Gatys et al. [1] and Luan et al. [5].)

2 Key Contributions and Main Ideas

The paper’s contributions can be summarized as follows:

1. **PhotoWCT Stylization:** The authors extend the WCT method [3] by modifying the autoencoder architecture. Standard WCT transfers style by aligning the second-order statistics of deep features, but when applied to photorealistic images it often introduces spatial artifacts. By replacing conventional upsampling with unpooling (using pooling masks), PhotoWCT preserves spatial structure and reduces distortions. (See Appendix B, Figure 3, for a visual comparison of WCT and PhotoWCT.)
2. **Graph-Based Smoothing:** Recognizing that the stylization output may exhibit inconsistencies in homogeneous regions, the paper introduces a smoothing step formulated as a closed-form quadratic optimization problem. By constructing an affinity matrix (typically computed using a matting affinity to capture local statistics) from the content image and solving the resulting sparse linear system, the method enforces spatial consistency and significantly enhances photorealism.

Together, these innovations yield a pipeline that is both computationally efficient and effective at preserving image structure.

3 Proposed Method

The overall framework comprises two primary stages: stylization with PhotoWCT and graph-based smoothing.

3.1 Stylization with PhotoWCT

The process begins by passing the content image I_C and style image I_S through a pretrained VGG-19 encoder to extract deep features. The goal is to transform the content features H_C so that their covariance matches that of the style features H_S . This is achieved through the Whitening and Coloring Transform (WCT), which computes the projection matrices

$$P_C = E_C \Lambda_C^{-\frac{1}{2}} E_C^T \quad \text{and} \quad P_S = E_S \Lambda_S^{\frac{1}{2}} E_S^T,$$

where E_C and E_S are the eigenvector matrices and Λ_C and Λ_S are the diagonal eigenvalue matrices obtained from the content and style feature covariances, respectively. The transformed content features are then computed as:

$$H_{CS} = P_S P_C H_C.$$

PhotoWCT further refines this transformation by using unpooling layers in the decoder instead of standard upsampling, which aids in recovering spatial details lost during max-pooling. Although semantic segmentation (for example, using a simple two-label mask for sky versus ground) is not always necessary, it can be applied selectively to improve results when the content and style images contain distinct semantic regions.

3.2 Graph-Based Smoothing via Laplacian Regularization

Even after stylization, the initial output Y may exhibit inconsistent textures. To enforce spatial consistency, the method refines Y by solving a quadratic optimization problem defined over a pixel affinity graph. Let y_i denote the color (or feature vector) of pixel i in Y and r_i denote the corresponding pixel value in the desired output R . The optimization is formulated as:

$$\min_R \frac{1}{2} \sum_{i,j} w_{ij} \left\| \frac{r_i}{\sqrt{d_{ii}}} - \frac{r_j}{\sqrt{d_{jj}}} \right\|^2 + \lambda \sum_i \|r_i - y_i\|^2, \quad (1)$$

where:

- w_{ij} is the affinity between pixels i and j , computed from the content image (using, for example, a matting affinity that accounts for local means and variances),
- $d_{ii} = \sum_j w_{ij}$ is the degree of pixel i ,
- λ is a regularization parameter.

Equation (1) enforces that neighboring pixels with high affinity have similar values in R while keeping R close to Y . By defining the normalized Laplacian matrix

$$S = D^{-1/2} W D^{-1/2},$$

and setting $\alpha = \frac{1}{1+\lambda}$, the closed-form solution is given by:

$$R^* = (1 - \alpha)(I - \alpha S)^{-1} Y.$$

This efficient solution is computed using sparse matrix factorization techniques, ensuring that the final output is both smooth and faithful to the initial stylization.

4 Optimization in Computer Vision

Optimization is the cornerstone of this approach, enabling both the feature transformation and spatial regularization in a computationally efficient manner.

4.1 Feature Transformation via Optimization

The Whitening and Coloring Transform can be interpreted as solving an optimization problem in deep feature space. By whitening the content features and then re-coloring them using the style features’ covariance, the method effectively aligns the statistical properties of the two images. The closed-form solution based on eigen-decomposition avoids iterative optimization and thereby improves processing speed.

4.2 Graph-Based Smoothing as Quadratic Optimization

The smoothing step, defined in Equation (1), uses Laplacian regularization to enforce that pixels with high affinity in the content image have similar values in the output. Solving this quadratic optimization problem in closed form (via the normalized Laplacian and sparse matrix inversion) yields a spatially consistent image. This approach is computationally efficient and supports real-time applications, which is a significant advantage over iterative methods.

4.3 Broader Impact

By integrating closed-form solutions in both stages, the proposed framework achieves a rare balance between quality and efficiency. The optimization techniques not only enhance the photorealism of the output but also open the door for real-time style transfer applications. This dual optimization paradigm represents a powerful advancement in computer vision, bridging the gap between artistic style transfer and practical, photorealistic image manipulation.

5 Experimental Evaluation and Critical Analysis

Our implementation was tested on two distinct content/style pairs, selected from online searches to represent novel data:

- **Set 1 (Desert/Green Hill):** In this challenging pair, the original stylization suffers from visible artifacts and a loss of fine texture, with the final output quality noticeably lower than that of the input images. However, by integrating smoothing in Lab space and employing semantic segmentation (using a simple two-label mask for sky vs. ground), the method significantly improves spatial consistency and preserves structural details. The Lab-space smoothing helps restore natural color transitions, while semantic segmentation guides the style transfer to appropriate regions.
- **Set 2 (Trees, Autumn vs. Spring):** For this pair, the overall photorealism is affected because the sky is not correctly recognized and is incorrectly rendered in an orange hue (typical of autumn), which diverges from the content image’s quality. Additionally, the quality of the final output is not as high as that of the original content and style images. Nevertheless, integrating Lab space in the smoothing stage does lead to noticeable improvements. In this case, blending the smoothed output with the original stylized image results in better preservation of fine textures.

It is important to note that the original implementation did not incorporate Lab-space smoothing or semantic segmentation. These modifications were introduced in our implementation to address shortcomings in the raw PhotoWCT output, particularly for challenging cases. While these enhancements improve photorealism, our experiments also reveal limitations—such as improper recognition of the sky in Set 2—indicating that further refinement (e.g., more sophisticated segmentation or adaptive blending) may be needed.

Overall, the experimental results underscore that while the proposed optimization-based framework is powerful, its performance is sensitive to the semantic content of the images. When the input images share similar semantic structures (as in Set 1), our modifications yield more photorealistic results. Conversely, when the content is more complex or misclassified (as in Set 2), even with Lab-space smoothing and blending, some artifacts may persist.

The transformation process is summarized as follows:



Figure 1: Two example sets used in our photorealistic style transfer experiments. **Top row:** Content (desert), Style (green hill), and Final Result. **Bottom row:** Content (spring trees), Style (autumn trees), and Final Result.

6 Conclusion

The paper introduces a novel, closed-form framework for photorealistic style transfer that overcomes many limitations of traditional neural style transfer methods. By decomposing the problem into a feature-level transformation (PhotoWCT) and a graph-based smoothing step, the method achieves significant gains in computational efficiency and visual quality.

Our implementation, which incorporates additional modifications not present in the original work—namely, smoothing in Lab space and the use of semantic segmentation—demonstrates the potential of these enhancements.

In summary, while our modifications (Lab-space smoothing and semantic segmentation) have enhanced the performance of the original method, particularly for Set 1, challenges remain for more complex scenes. These findings highlight both the promise of the closed-form optimization approach and the need for further research into adaptive segmentation and blending strategies for robust photorealistic style transfer.

References

- [1] Leon A Gatys, Alexander S Ecker, and Matthias Bethge. Texture synthesis using convolutional neural networks. In *Advances in Neural Information Processing Systems*, pages 262–270, 2015.
- [2] Leon A Gatys, Alexander S Ecker, and Matthias Bethge. Image style transfer using convolutional neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2414–2423, 2016.
- [3] Yijun Li, Chuan Fang, Jing Yang, Zhaowen Wang, Xia Lu, and Ming-Hsuan Yang. Universal style transfer via feature transforms. In *Advances in Neural Information Processing Systems*, pages 386–395, 2017.
- [4] Yijun Li, Ming-Yu Liu, Xueting Li, Ming-Hsuan Yang, and Jan Kautz. A closed-form solution to photorealistic image stylization, 2018.
- [5] F. Luan, S. Paris, E. Shechtman, and K. Bala. Deep photo style transfer. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 371–379, 2017.

A Figure comparing different methods (Gatys et al., Luan et al.)

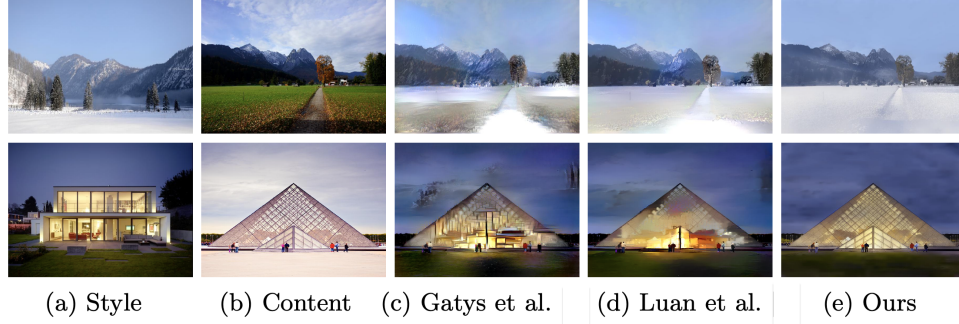


Figure 2: Given a style image (a) and a content image (b), our method transfers the style onto the content image, as illustrated in panels (c), (d), and (e). Compared with existing methods, the results are more consistent with fewer artifacts, and the processing is significantly faster.

B Figure comparing WCT vs. PhotoWCT

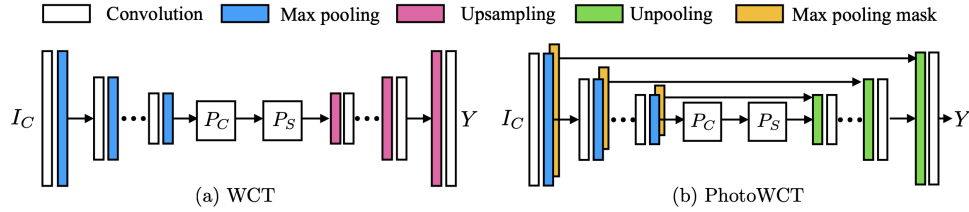


Figure 3: Both PhotoWCT and WCT use the same encoder architecture and projection steps. In PhotoWCT, however, conventional upsampling (shown in pink) is replaced by unpooling layers (in green), which, together with the pooling mask (yellow) that records the locations of maximum activations during pooling, help recover fine spatial details